

NAPAS, Russian Federation

WMO: 290230

Lat: 59.850N

Lon: 81.950E

Elev: 76

StdP: 100.42

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.8	(c) -38.5	(d) -44.1	(e) 0.0	(f) -41.7	(g) -40.9	(h) 0.1	(i) -38.3	(j) 6.9	(k) -12.6	(l) 5.9	(m) -12.4	(n) 0.9	(o) 30	(p) 0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	28.4	18.9	26.6	18.3	24.9	17.5	20.6	25.8	19.5	24.6	18.4	23.2	2.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	22.9	17.6	12.7	21.7	16.6	11.9	20.7	59.8	25.8	56.0	24.7	52.7	23.2	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	5.0	-43.6	31.1	3.9	1.6	-46.5	32.3	-48.7	33.3	-50.9	34.2	-53.8	35.4		
			-43.0	22.1	3.3	1.5	-45.4	23.2	-47.3	24.1	-49.2	24.9	-51.6	26.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-21.6	-17.5	-8.7	-1.1	7.4	16.0	17.8	14.0	8.0	-0.1	-12.2	-19.4
	DBStd	15.25	9.25	7.99	6.75	6.19	6.13	4.65	3.70	3.70	4.49	5.51	9.39	9.65
	HDD10.0	4789	979	771	581	332	127	7	0	7	90	316	667	912
	HDD18.3	7269	1238	1004	839	582	342	98	55	141	309	573	917	1170
	CDD10.0	640	0	0	0	1	46	186	243	132	31	1	0	0
	CDD18.3	77	0	0	0	0	3	27	40	7	0	0	0	0
	CDH23.3	716	0	0	0	0	44	257	339	72	4	0	0	0
	CDH26.7	147	0	0	0	0	9	53	73	12	0	0	0	0
(14) Wind	WSAvg	2.2	2.0	2.2	2.6	2.7	2.7	2.2	1.8	1.9	2.0	2.4	2.3	2.1
(15) Precipitation	PrecAvg	607	28	23	27	33	59	71	79	82	60	54	51	40
	PrecMax	793	62	50	59	77	133	165	148	152	133	98	78	85
	PrecMin	421	5	3	6	3	18	2	21	15	6	23	28	14
	PrecStd	95	14	10	13	15	29	34	35	39	27	17	16	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	0.5	7.2	16.2	27.9	30.2	31.2	28.5	23.6	14.7	3.8	0.2
		MCWB	-0.7	-0.4	2.6	8.8	16.1	18.6	21.0	19.9	16.6	10.1	2.6	-0.3
	2%	DB	-3.1	-1.7	4.3	12.6	23.7	28.3	28.6	25.5	20.6	11.2	1.7	-2.0
		MCWB	-3.6	-2.5	1.0	7.0	14.3	18.2	19.9	18.1	14.7	8.4	1.1	-2.6
	5%	DB	-6.1	-4.1	2.2	10.0	20.5	26.5	26.8	23.2	18.0	9.2	0.7	-4.0
		MCWB	-6.8	-4.9	-0.2	5.6	12.8	17.6	19.1	17.4	13.3	7.1	0.2	-4.6
	10%	DB	-8.8	-6.6	0.5	7.6	17.2	24.4	25.1	20.8	15.5	7.1	-0.4	-6.1
		MCWB	-9.4	-7.5	-1.2	3.8	11.4	16.9	18.2	16.2	12.1	5.6	-1.0	-6.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.2	3.1	10.0	17.0	21.2	23.1	20.8	17.5	10.7	2.7	-0.2
		MCDB	-0.5	0.3	6.5	14.3	25.1	26.7	28.7	26.0	22.2	13.3	3.3	0.1
	2%	WB	-3.6	-2.4	1.3	7.6	15.1	19.6	21.1	19.2	15.3	8.9	1.2	-2.5
		MCDB	-3.1	-1.6	3.4	12.0	22.6	25.2	26.3	23.9	19.5	11.0	1.5	-2.0
	5%	WB	-6.6	-4.9	0.1	5.8	13.5	18.6	20.1	17.9	13.8	7.3	0.3	-4.5
		MCDB	-6.0	-4.1	1.9	9.4	19.2	24.1	25.3	22.0	17.3	9.0	0.8	-4.0
	10%	WB	-9.3	-7.4	-1.2	4.1	11.7	17.7	19.1	16.7	12.3	5.5	-1.0	-6.6
		MCDB	-8.8	-6.6	0.6	7.3	17.0	23.2	23.8	20.1	14.9	6.9	-0.4	-6.1
(35) Mean Daily Temperature Range	MDBR		8.1	9.5	10.7	10.1	9.6	9.8	9.6	8.9	8.0	5.8	6.9	7.7
	5% DB	MCDBR	8.2	7.7	10.6	12.5	14.3	13.4	12.5	12.7	13.0	8.9	5.2	8.9
		MCWBR	8.2	7.4	8.1	8.0	6.9	5.3	5.3	6.4	7.2	6.0	5.0	8.5
	5% WB	MCDBR	8.2	7.6	9.7	11.2	13.0	11.5	11.4	11.2	11.5	8.2	5.1	8.9
		MCWBR	8.1	7.4	7.7	7.6	6.8	5.3	5.4	6.3	6.8	5.9	4.9	8.6
(40) Clear-Sky Solar Irradiance	taub		0.268	0.304	0.353	0.375	0.387	0.403	0.447	0.411	0.361	0.329	0.278	0.261
	taud		1.970	2.003	1.951	2.053	2.278	2.322	2.172	2.264	2.363	2.263	2.023	1.913
	Ebn at Noon		463	660	742	811	838	829	774	775	758	646	445	308
	Edh at Noon		51	87	126	137	118	116	131	111	85	67	47	36
(44) All-Sky Solar Radiation	RadAvg		0.49	1.42	2.94	4.47	4.90	5.64	5.09	3.73	2.31	1.08	0.48	0.26
	RadStd		0.05	0.12	0.19	0.44	0.56	0.56	0.54	0.45	0.30	0.11	0.10	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page